



GPT-2

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The Dimensionality of Transformers

Unveiling GPT-2: Reconstructing the Geometry of the Embedding Space

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STUDY
ANHOLL
RTIOALM
SPACE

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VECTORIAL
HIGH SPACE
VECTORIAL
SPACE

1. Theoretical overview
2. Aims of the project
3. Technical approach
4. Results
5. Conclusions

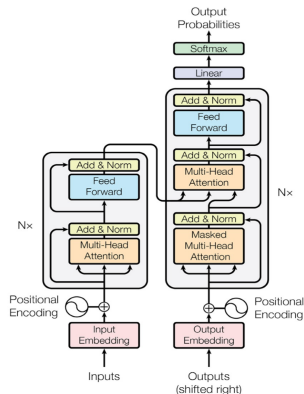
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Transformer internal structure

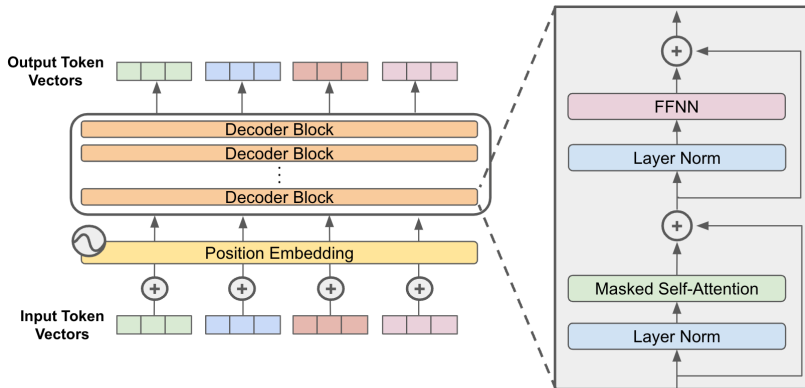


For each decoder

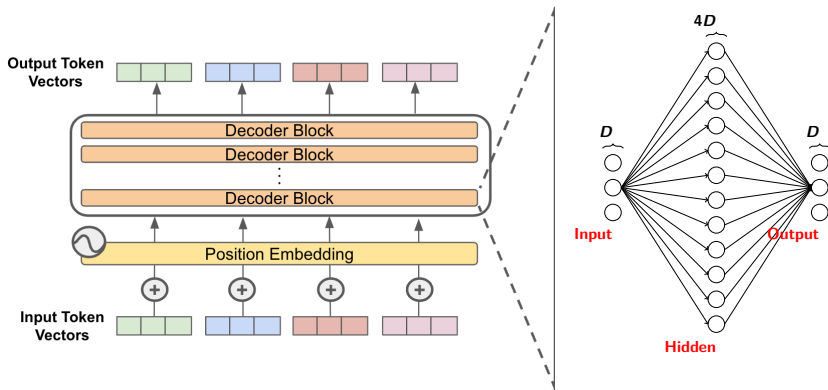
1. Positional embedding
converting words to tokens and tokens to vectors
2. Attention Mechanism
with twelve Attention Heads to capture the context
3. Addition & normalization
to connect and stabilize each layer
4. Feed Forward Neural Network
with one hidden layer



Transformer internal structure



Transformer internal structure



The attention mechanism



- Simulates human attention by assigning varying **degrees of correlation** among words in a sentence



- The dependencies can be related to semantic meaning and **context**
- Enhances the transformer's ability to capture **long-range dependencies**

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with matrices and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representations and became the other person involved in nearly every detail. Niki designed, implemented, tested and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukas and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

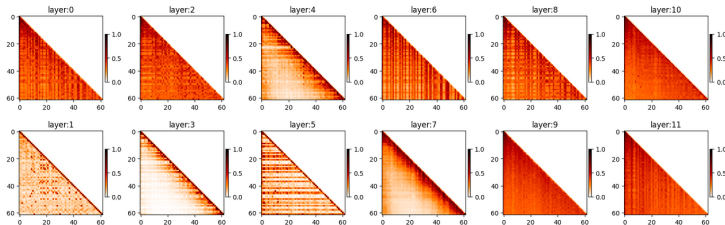
*Work performed while at Google Brain.

*Work performed while at Google Research.

The attention mechanism

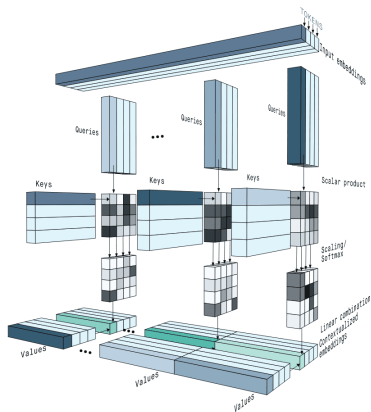


- The weights for the vectors in the embedding space are calculated by the **twelve attention heads**

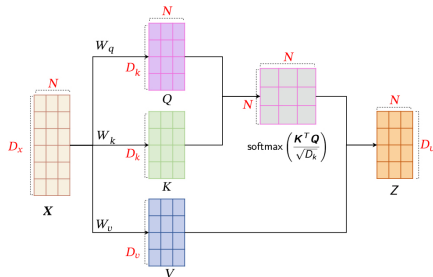
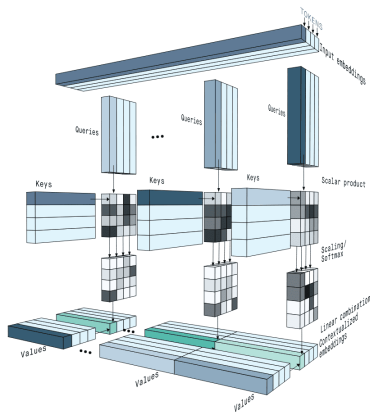


- Contextual **proximity matrices** are built and dynamically adjusted

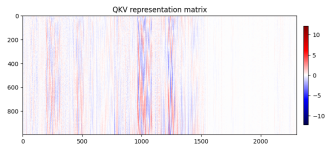
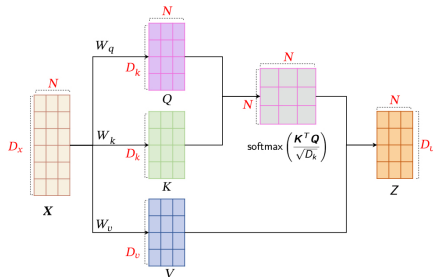
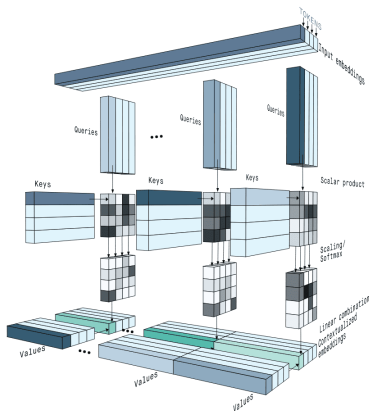
The attention mechanism



The attention mechanism



The attention mechanism



From one of our prompts

1. Theoretical overview
2. Aims of the project
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Aims of the project

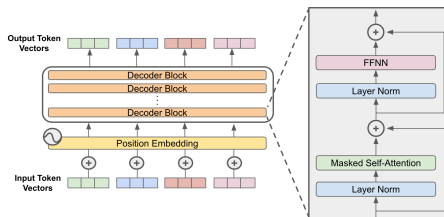


- Opening the **transformer black box**

What exactly happens in there?

How is the decoder structure layered?

How is the prompt processed?

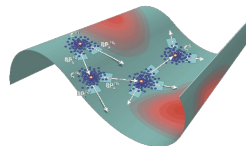
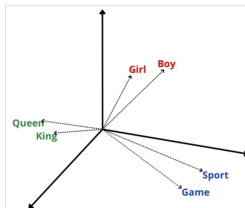


Aims of the project



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- Opening the **transformer black box**
What exactly happens in there?
How is the decoder structure layered?
How is the prompt processed?
- Inspecting the **embedding space**
How does the geometry of the embedding space evolve?
How does its dimensionality evolve?



- Opening the **transformer black box**
What exactly happens in there?
How is the decoder structure layered?
How is the prompt processed?
- Inspecting the **embedding space**
How does the geometry of the embedding space evolve?
How does its dimensionality evolve?
- Studying the mechanism behind the **last token** prediction
How does the model produce each next-token output?
How do the heads express the last token's dependencies on previous ones?

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The generative model we used is **GPT-2 small**

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117 million parameters

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117 million parameters

Our starting point: use of prompts with different **lengths** and **semantic areas**

1. **Pre-processing**

Encoding information inside the model

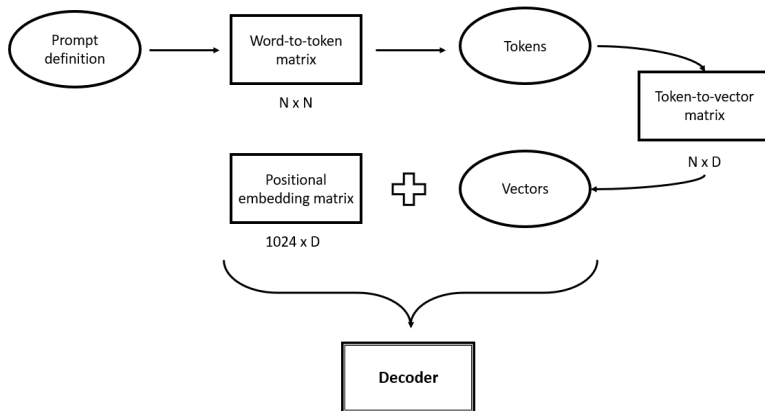
2. Study of the **dimensionality** of the embedding space

Observing how the intrinsic dimension changes over the process

Computing the relative distances between elements in the space, determining an appropriate metric

3. **Last-token analysis**

The prediction of the last token depends only on the result of the last token in the prompt

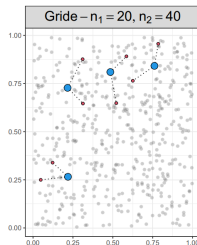


Approaches:

- Basic clustering with **t-SNE**
- Dadapy** {
 - Dimensional analysis
 - Metrics comparison
- **PCA** with SVD
- Manifold volume with **Gram matrix**

- Generalized Ratios Intrinsic Dimension Estimator: statistical method introduced in the DADApY package
- Allows **estimating the intrinsic dimension (ID) of a manifold** as an explicit function of the scale of the distances among data points

- Generalized Ratios Intrinsic Dimension Estimator: statistical method introduced in the DADApY package
- Allows **estimating the intrinsic dimension (ID) of a manifold** as an explicit function of the scale of the distances among data points
- Method:
 1. choose two nearest neighbors: the n_1 -th and the n_2 -th (typically $n_1 = 2n_2$)
 2. exploit the maximum log-likelihood properties to estimate the ID based on n_1 , n_2 and the radii of the correspondent hyperspheres of data points



Reference: <https://doi.org/10.1038/s41598-022-20991-1>

Prompts selection

Various **prompt sizes**

1. 50 words
2. 300 words
3. 950 words

Different **semantic area** and **text structure**

1. List
2. Poetry
3. Instructions
4. Plain text

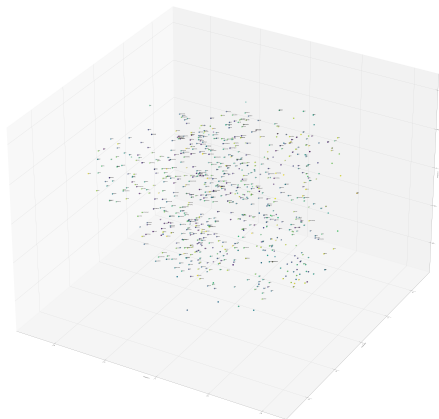
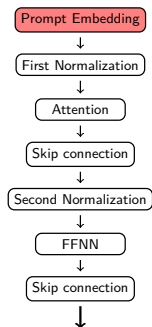
	List	Poetry	Instructions	Text 950	Text 300	Text 50
Nature	trees forest river mountains wildlife ecosyste...	In nature's realm, where wonders bloom, A tape...	Taking care of biodiversity in a national park...	Nature, in its vast expanse and intricate deta...	Nature, with its breathtaking beauty and intri...	Nature encompasses Earth's landscapes, ecosyst...
Astronomy and Physics	astronomy physics astrophysics cosmology unive...	In the velvet expanse where stars ignite, Astr...	Studying the phenomenon of dark energy and gal...	Astronomy and physics are two intertwined fiel...	Astronomy and physics stand as pillars of scie...	Astronomy explores the vastness of the univers...
Mathematics	addition subtraction multiplication division a...	In the realm where numbers weave their dance, ...	Solving differential equations involves a syst...	Mathematics is the language of patterns, struc...	Mathematics is the universal language of patte...	Mathematics is the language of patterns, shape...
Psychology	psychology cognition perception memory learnin...	In minds' depths, thoughts weave and sway, Emo...	Providing psychotherapy involves a structured ...	Psychology, as a field of study, delves into t...	Psychology is the scientific study of the mind...	Psychology delves into the complexities of the...
Music	melody harmony rhythm beat tempo dynamics pitc...	In the silence before the first note's birth, ...	Composing a musical piece is a creative endeav...	Music, a universal language, transcends cultur...	Music is a universal language that transcends ...	Music is a universal language that transcends ...

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Semantic cluster visualization

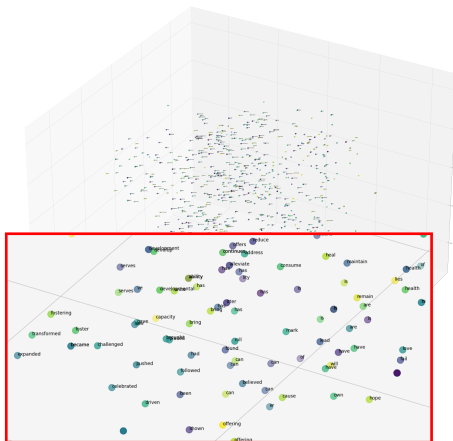


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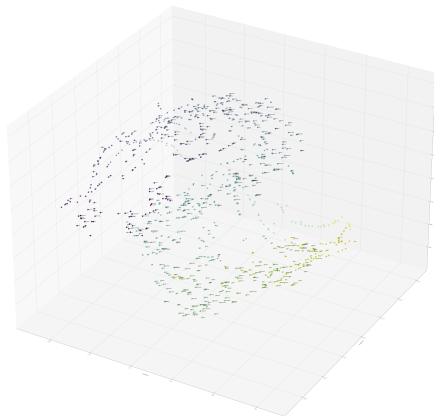
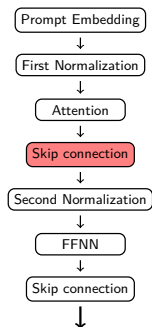



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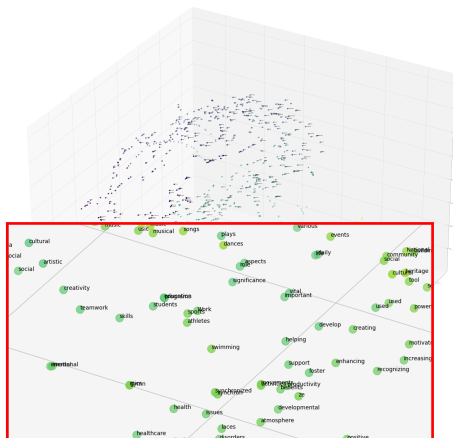
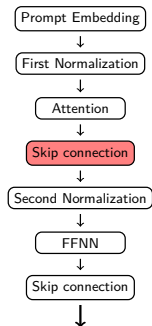
graph TD
    A[Prompt Embedding] --> B[First Normalization]
    B --> C[Attention]
    C --> D[Skip connection]
    D --> E[Second Normalization]
    E --> F[FFNN]
    F --> G[Skip connection]
    G --> H[ ]
  
```



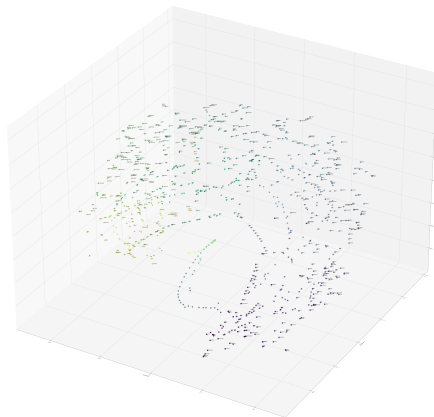
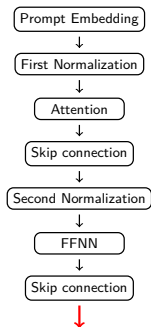
Semantic cluster visualization



Semantic cluster visualization

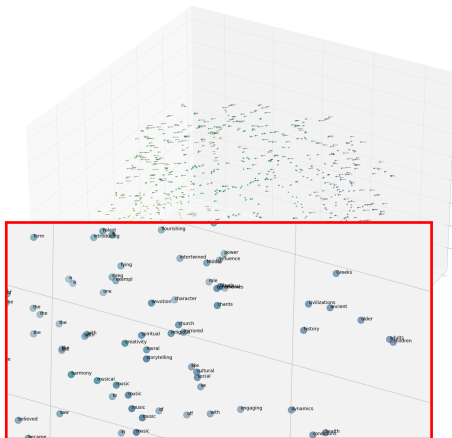


Semantic cluster visualization

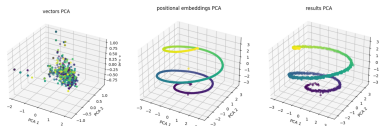



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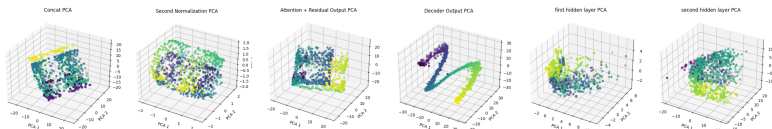
graph TD
    A[Prompt Embedding] --> B[First Normalization]
    B --> C[Attention]
    C --> D[Skip connection]
    D --> E[Second Normalization]
    E --> F[FFNN]
    F --> G[Skip connection]
    G --> H[Output]
    style H fill:#f00,stroke:#f00,stroke-width:2px
  
```



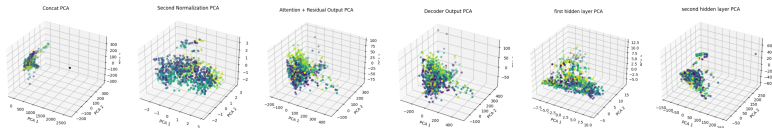
Positional Embedding:



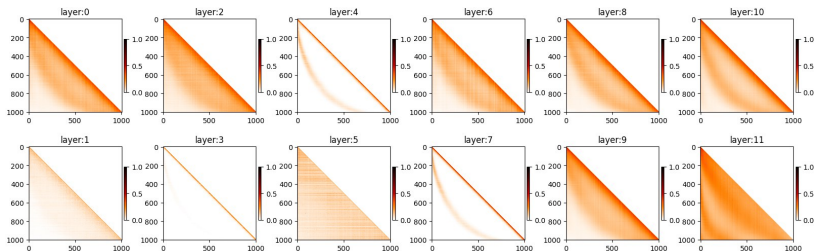
First Decoder:



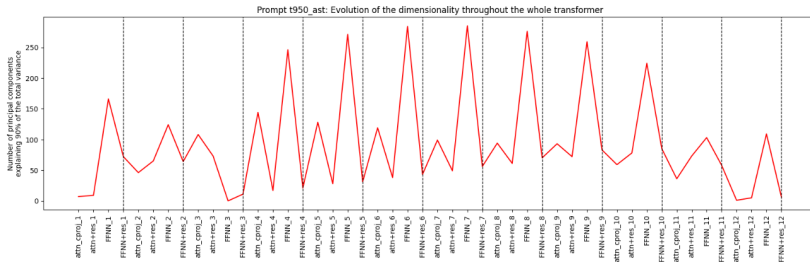
Last Decoder:



QKV matrices and attention heads



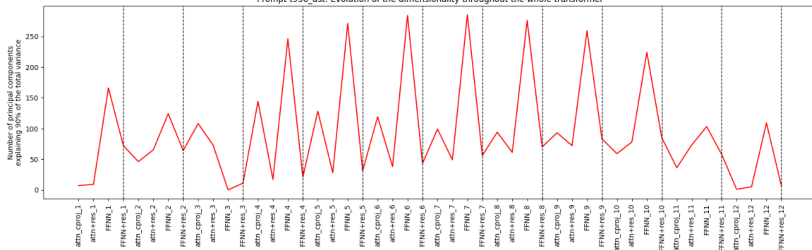
Dimensionality evolution



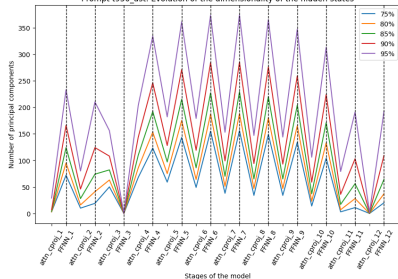
Dimensionality evolution



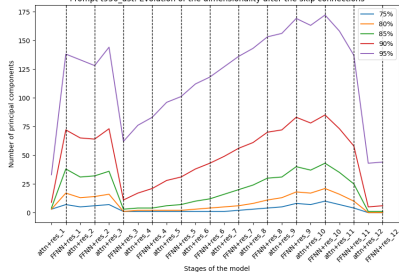
Prompt t950_ast: Evolution of the dimensionality throughout the whole transformer



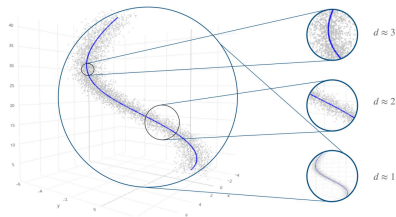
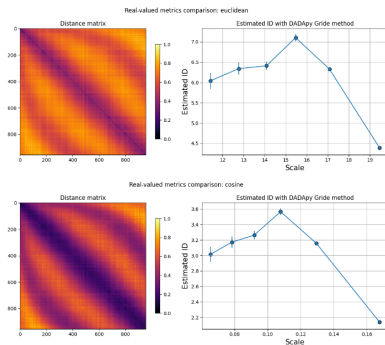
Prompt t950_ast: Evolution of the dimensionality of the hidden states



Prompt t950_ast: Evolution of the dimensionality after the skip connections



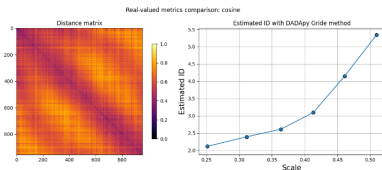
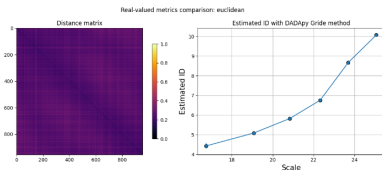
1000-token prompt First decoder



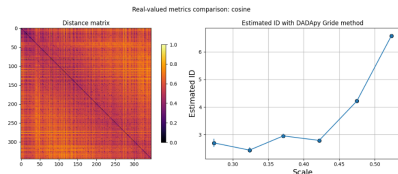
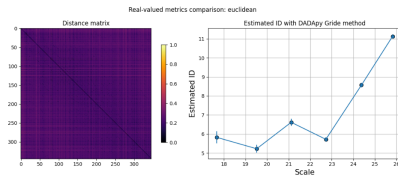
Intrinsic Dimension and Metrics



1000-token prompt



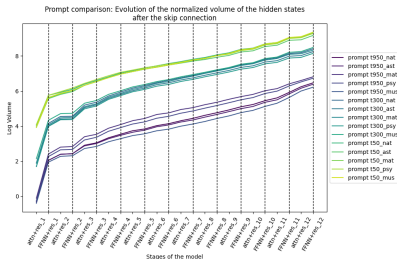
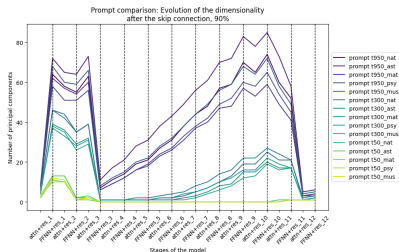
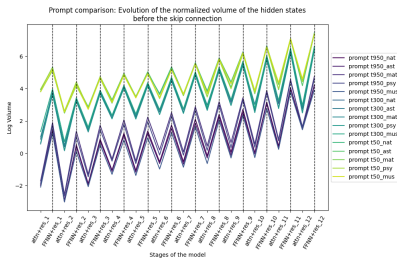
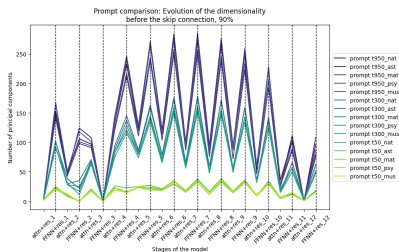
300-token prompt



Statistical sample analysis



Prompt size comparison: [955, 980, 863, 889, 1003, 405, 434, 368, 409, 345, 63, 66, 58, 54, 62]

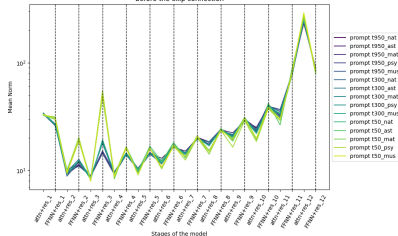


Statistical sample analysis

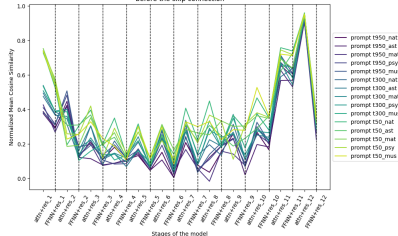


Prompt size comparison: [955, 980, 863, 889, 1003, 405, 434, 368, 409, 345, 63, 66, 58, 54, 62]

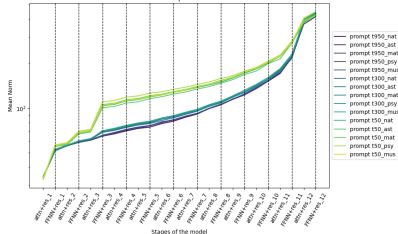
Prompt comparison: Evolution of the mean norm of the hidden states before the skip connection



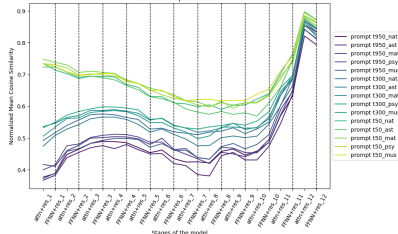
Prompt comparison: Evolution of the mean cosine similarity before the skip connection



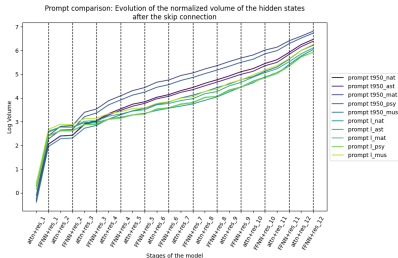
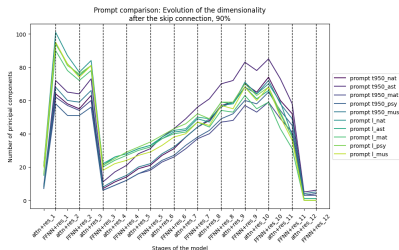
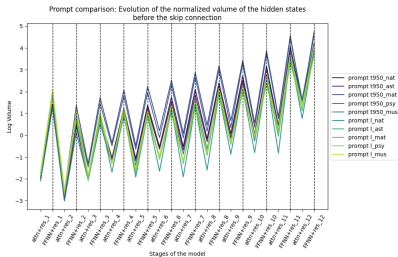
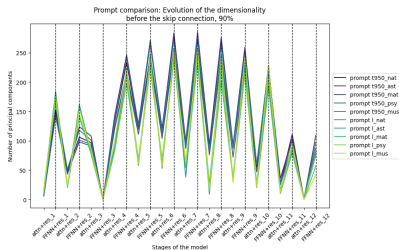
Prompt comparison: Evolution of the mean norm of the hidden states after the skip connection



Prompt comparison: Evolution of the mean cosine similarity after the skip connection

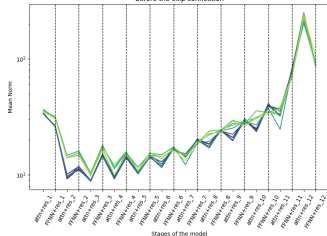


Prompt type comparison:

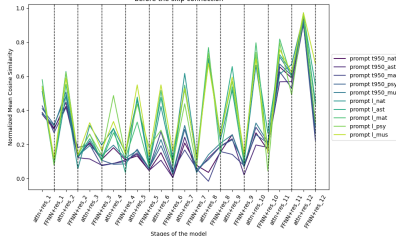


Prompt type comparison:

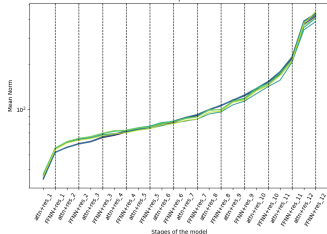
Prompt comparison: Evolution of the mean norm of the hidden states before the skip connection



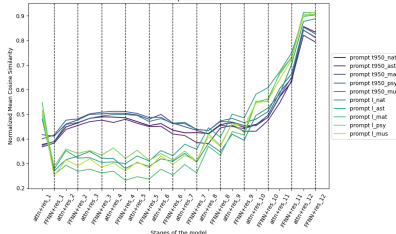
Prompt comparison: Evolution of the mean cosine similarity before the skip connection



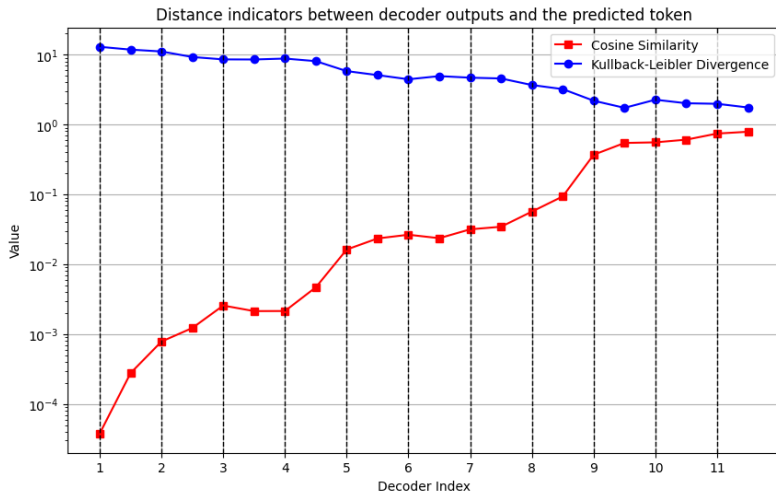
Prompt comparison: Evolution of the mean norm of the hidden states after the skip connection



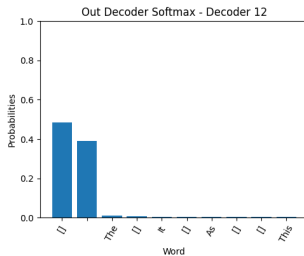
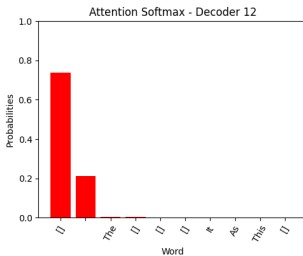
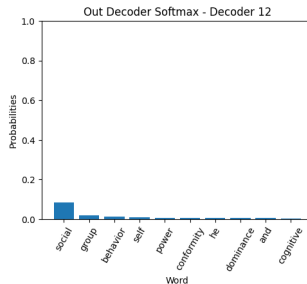
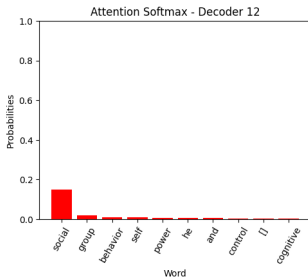
Prompt comparison: Evolution of the mean cosine similarity after the skip connection



Last token analysis



Last token analysis



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- Opening the **transformer black box**
 - t-SNE and PCA plots show how the prompt is processed, layer-by-layer and decoder-by-decoder
- Inspecting the **embedding space**
 - Metrics and ID plots, the vectors are on a low-dim manifold (t-SNE)
 - Evolution of relevant quantities with statistical approach
- Studying the mechanism behind the **last token** prediction
 - GIF of the probability distribution evolution
 - Attention heads heatmaps express inter-dependencies between tokens



GPT-2

Thank you.

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SPACE

SPACE
VECTORIAL
HIGH SPACE
VECTORIAL
SPACE